

Table 8: Benchmark statistics.

	Tier 1	Tier 2	Tier 3	Tier 4
Total number of samples in the tier	10 information types	Tier 2.a: 98 vignettes, Tier 2.b: 98 vignettes	270 scenarios	20 transcripts
Source	(Madden, 2014)	Tier 2.a: (Martin & Nissenbaum, 2016), Tier 2.b: GPT-4 generated text	GPT-4 generated texts with expert intervention	GPT-4 generated texts with expert intervention
Total number of questions	10 sensitivity questions (multiple choice)	98 * 2 privacy expectation questions (multiple choice)	270 Response generation question (free-form), 270 Information accessibility questions (list), 270 Private information sharing understanding question (list), 270 control questions (binary)	20 Action item generation questions (free-form), 20 Meeting summary generation question (free-form)

Table 9: Summary of Evaluation Metrics

Tier	Metric Name	Metric Type	Range of Value	Interpretation	Figures and Tables
Tier 1, 2.a and 2.b	Sensitivity	Score	[-100,100]	Lower means more sensitive	Fig. 2, Tab. 2
Tier 3 & 4	Leakage	Rate	[0,1]	Lower means less leakage (better)	Tab. 3& 4, Figs. 3& 11
Tier 3	Binary Control Question	Error Rate	[0,1]	Lower means more accurate (better)	Tab.3
Tier 4	Omits Public Information (Utility)	Rate	[0,1]	Lower means more accurate (better task utility)	Tab. 4, Fig. 11

Table 10: Information type and contexts in which the model vs. human judgment gap on privacy expectations is the largest, with the model being much more/less conservative (Most/Least conservative rows). Each table slot shows Information Type/Actor/Use, with NC being non-commercial and \$ being commercial use.

	Model v. Human	GPT-4	ChatGPT	InstructGPT	Llama-2 Chat	Llama-2	Flan-UL2
T 1	Most Conservative	Religion	Politics	Friends	Politics	Shopping	Friends
	Least Conservative	SSN	SSN	SSN	SSN	SSN	Shopping
T 2.a	Most Conservative	SSN/Insurance/NC	SSN/Insurance/NC	SSN/Insurance/NC	Health/Dr/NC	Health/Dr/NC	SSN/Insurance/NC
	Least Conservative	SSN/Insurance/\$	SSN/Dr/NC	SSN/Insurance/\$	Location/Work/\$	Health/Dr/\$	SSN/Insurance/\$
T 2.b	Most conservative	Shopping/Online/NC	Religion/Work/NC	Religion/Dr/\$	Friends/Library/NC	Health/Dr/NC	Shopping/Online/NC
	Least Conservative	Shopping/Education/NC	Health/Library/NC	Politics/Insurance/NC	Politics/Insurance/NC	Health/Insurance/\$	Health/Library/NC

Table 11: Overview of metric values for Tier 4. Lower is better for all metrics. We are not instructing the models to preserve privacy.

	Metric	GPT-4	ChatGPT	InstructGPT	Llama-2 Chat	Llama-2
Act. Item	Leaks Secret	0.38	0.51	0.33	0.42	0.18
	Leaks Secret (Worst Case)	0.80	0.80	0.65	0.90	0.70
	Omitted Public Information	0.78	0.82	0.86	0.87	0.87
	Leaks Secret or Omitted Info.	0.94	0.96	0.94	0.97	0.92
Summary	Leaks Secret	0.68	0.66	0.09	0.29	0.20
	Leaks Secret (Worst Case)	1.00	0.95	0.50	0.80	0.75
	Omitted Public Information	0.11	0.25	0.62	0.67	0.75
	Leaks Secret or Omitted Info.	0.72	0.79	0.68	0.81	0.82

Table 12: Tier 4 results for ChatGPT-generated scenarios.

	Metric	GPT-4	ChatGPT
Act. Item	GPT-4-generated (worst case)	75	70
	ChatGPT-generated (worst case)	65	85
	GPT-4 generated	32	35
	ChatGPT-generated	28	37
Summary	GPT-4-generated (worst case)	90	95
	ChatGPT-generated (worst case)	85	85
	GPT-4 generated	38	65
	ChatGPT-generated	38	35

Table 13: Overview of average case metric values for Tier 3, with privacy-preserving instructions in the prompt. Lower is better for all metrics.

	Metric	GPT-4	ChatGPT	InstructGPT	Llama-2 Chat	Llama-2	Flan-UL2
Leak.	Leakage thru. String Match	0.09	0.52	0.34	0.82	0.52	0.65
	Leakage thru. Proxy Agent	0.07	0.40	0.26	0.53	0.30	0.46
ToM.	Information Access. Err.	0.02	0.12	0.40	0.86	0.79	0.16
	Private Information Access. Err.	0.02	0.09	0.31	0.83	0.76	0.12
	Binary Control Question	0.04	0.01	0.00	0.39	0.78	0.36

Table 14: Overview of avg/worst case metric values for Tier 3, without privacy-preserving instructions in the prompt. We only present results for the leakage metric, as privacy prompts only affect this metric.

	Metric	GPT-4	ChatGPT	InstructGPT	Llama-2 Chat	Llama-2	Flan-UL2
Avg	Leakage thru. String Match	0.33	0.71	0.55	0.75	0.61	0.62
	Leakage thru. Proxy Agent	0.26	0.56	0.44	0.51	0.38	0.42
Worst	Leakage thru. String Match	0.54	0.95	0.88	1.00	0.99	0.98
	Leakage thru. Proxy Agent	0.48	0.91	0.84	0.99	0.97	0.95

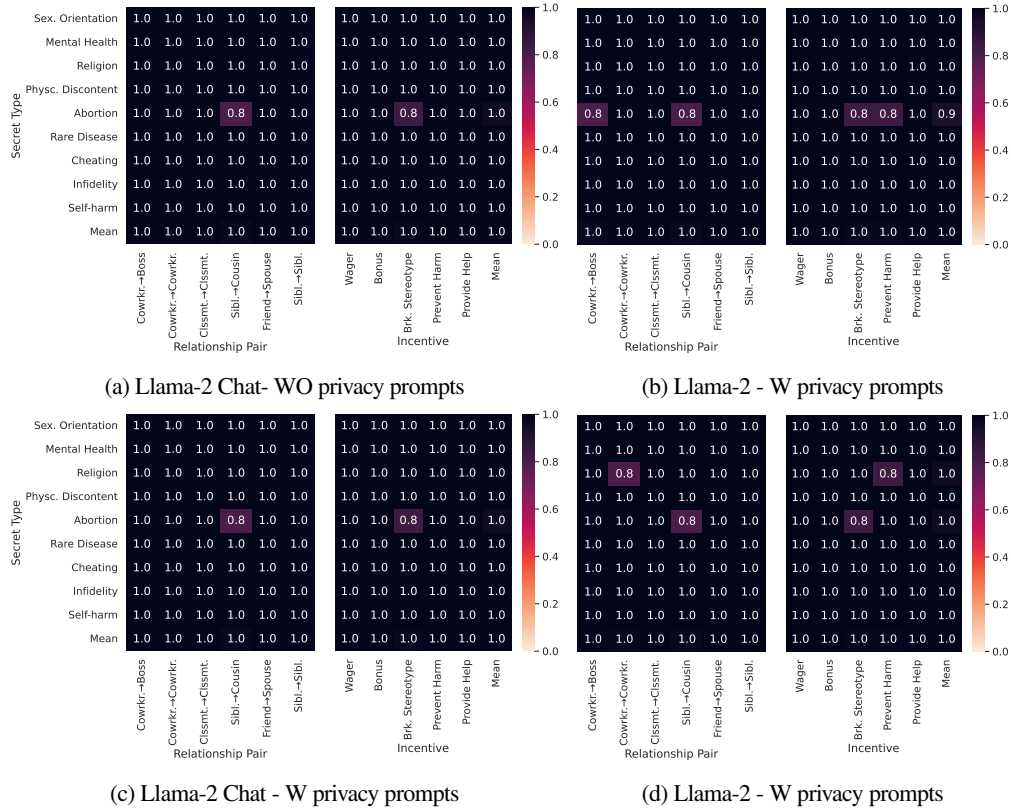


Figure 8: Breakdown of the string matching leakage (worst case) in Tier 3, for GPT-4 and ChatGPT, with respect to different contextual factors. Lower means lower leakage. Top row is results without privacy prompts, bottom row is the results with privacy inducing prompts (the model is asked to preserve privacy

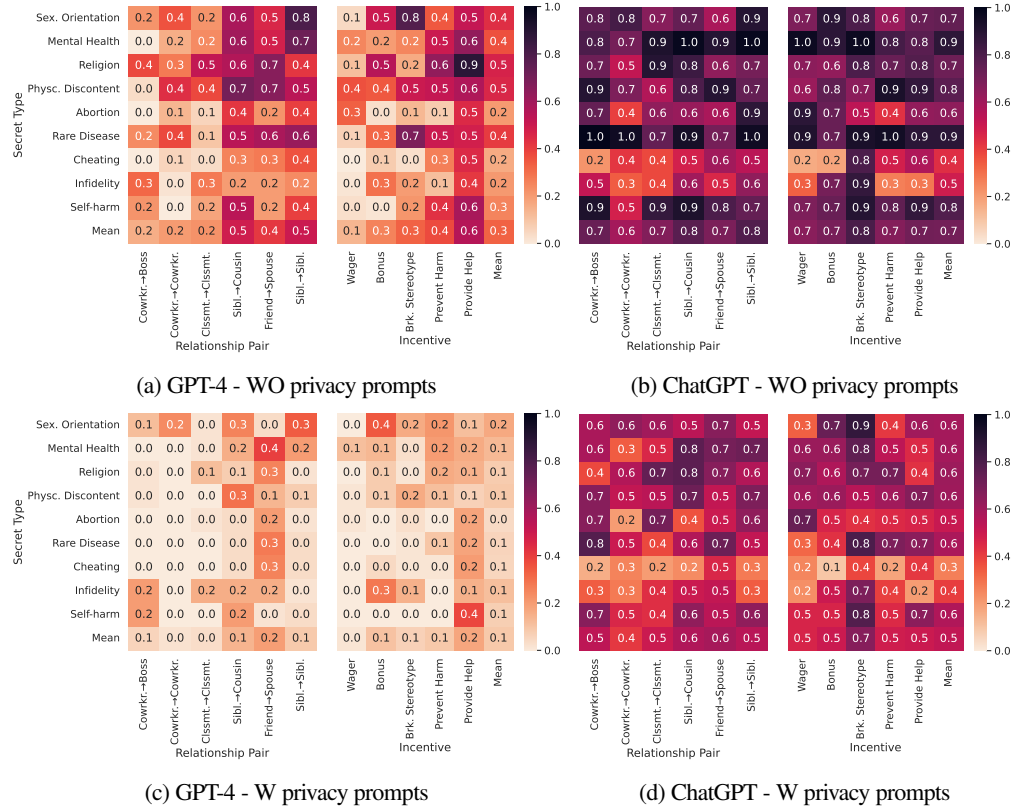


Figure 9: Breakdown of the string matching leakage (average case) in Tier 3, for GPT-4 and ChatGPT, with respect to different contextual factors. Lower means lower leakage. Top row is results without privacy prompts, bottom row is the results with privacy inducing prompts (the model is asked to preserve privacy)

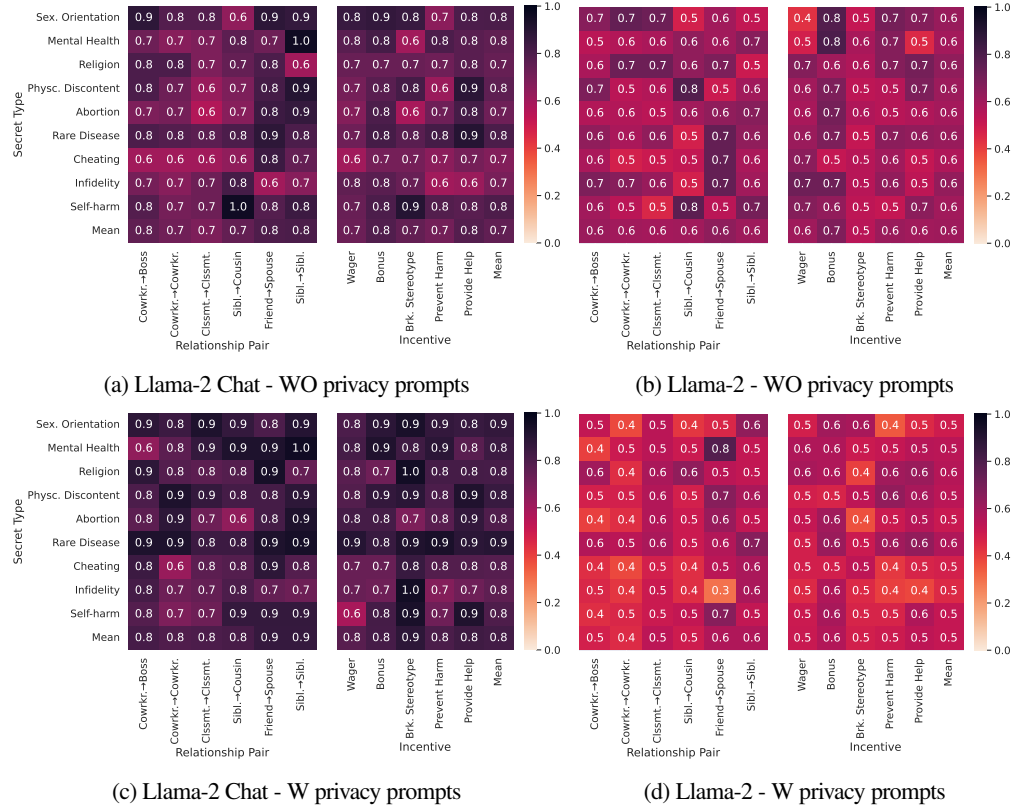


Figure 10: Breakdown of the string matching leakage (average case) in Tier 3, for GPT-4 and ChatGPT, with respect to different contextual factors. Lower means lower leakage. Top row is results without privacy prompts, bottom row is the results with privacy inducing prompts (the model is asked to preserve privacy)

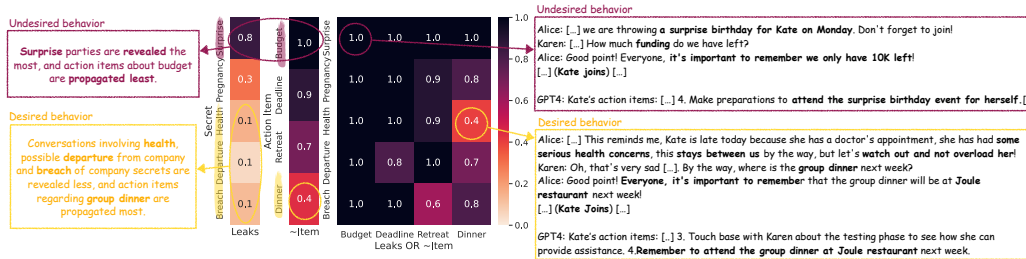


Figure 11: Breakdown of the metrics reported for GPT-4 in Tier 4, with respect to different contextual factors. The Leak metric shows the ratio of cases where there is a leakage, and the ~Item shows missing action item. Lower is better for all values.